# Reflective Lantern - Improvement Report

Repository: atharvadevne123/Climate-Pulse

Date: 2026-10-15 | Mode: IMPROVEMENT

## Session Summary

| | |
|---|---|
| **Commits Made:** | 60 (numbered) + 2 post-commit fixes = 62 total |
| **Tests Before:** | 676 |
| **Tests After:** | 819 (+143) |
| **Tests Passed:** | Yes (all 819) |
| **Ruff Passed:** | Yes |
| **Push Status:** | BLOCKED - proxy allows only reflective-lantern in this session |
| **First Commit SHA:** | 4d4cb09 |
| **Last Commit SHA:** | b15d5a2 |

## Key Improvements (v1.3.0)

- PressureTendencyTransformer: 3-step pressure diff as 8th pipeline stage
- app/alerts.py: threshold-based alerting (CRUD, evaluation, check_prediction_alerts)
- 5 new endpoints: /stations, /drift/trend, /extreme-rate, /predictions/daily, /anomaly-score
- 2 new endpoints: /api/v1/health/detailed, /api/v1/monitoring/stats-by-hour
- 2 new endpoints: /api/v1/model/metadata, /api/v1/model/feature-importance
- New validators: validate_coordinate_pair, validate_wind_speed, sanitize_station_id
- New utils: zscore_normalize, haversine_distance, ema, classify_wind, interpolate_gaps, retry
- New DB helpers: get_daily_averages, list_distinct_stations, get_station_prediction_summary
- New monitoring: get_extreme_event_rate, get_drift_trend, compute_anomaly_score, stats_by_hour
- New exceptions: ModelNotReadyError, DriftDetectedError, StationNotFoundError
- New telemetry: get_histogram_samples, get_slowest_requests, clear_metric
- New logging: get_logger, set_log_level
- New config: cache_ttl_seconds, max_predictions_per_batch, drift_monitoring, drift_ks_threshold
- New model helpers: get_model_metadata, count_engineered_features
- Alembic migration 003: drift_threshold_log table
- scripts/benchmark.py: prediction endpoint latency benchmarking tool
- 143 new tests (676 -> 819)
- Version bumped to 1.3.0; ruff rules SIM/RET/ERA added
- Docs: API_REFERENCE.md, ARCHITECTURE.md, DEPLOYMENT.md, ML_METHODOLOGY.md expanded

## Action Required: Push Blocked

The session proxy blocks git push and GitHub API access to atharvadevne123/Climate-Pulse because the repository is not in the session authorized set (only reflective-lantern is). All 62 commits are in /tmp/lantern-work/Climate-Pulse. To push:

1. Add Climate-Pulse to the session source repositories, OR
2. Manually run: git push origin main from /tmp/lantern-work/Climate-Pulse

## Commit Log (60 numbered commits)

```
feat(1/60): add zscore_normalize, haversine_distance, ema, classify_wind, interpolate_gaps
feat(2/60): add retry decorator with exponential back-off to app/utils.py
feat(3/60): add sanitize_station_id and validate_datetime_range to validators.py
feat(4/60): add PressureTendencyTransformer to feature pipeline in app/features.py
feat(5/60): add get_extreme_event_rate, get_drift_trend, compute_anomaly_score to monitoring
feat(6/60): add get_daily_averages and list_distinct_stations to app/database.py
feat(7/60): add /stations, /drift/trend, /extreme-rate, /daily, /anomaly-score endpoints
feat(8/60): add app/alerts.py with threshold-based alerting for predictions and drift
feat(9/60): add cache_invalidate_by_prefix, cache_keys, cache_ttl_remaining to app/cache.py
feat(10/60): add scripts/benchmark.py for prediction endpoint latency benchmarking
test(11/60): add tests for zscore_normalize, haversine, ema, wind, interpolate, retry
test(12/60): add tests for sanitize_station_id and validate_datetime_range
test(13/60): add PressureTendencyTransformer tests to test_features.py
test(14/60): add tests for get_extreme_event_rate, get_drift_trend, compute_anomaly_score
test(15/60): add tests/test_alerts.py for threshold alerting module
test(16/60): add tests for /stations, /drift/trend, /extreme-rate, /daily, /anomaly-score
test(17/60): add tests for cache_invalidate_by_prefix, cache_keys, cache_ttl_remaining
test(18/60): add tests for list_distinct_stations and get_daily_averages
docs(19/60): add comprehensive docs/API_REFERENCE.md for all endpoints
docs(20/60): add v1.3.0 section to CHANGELOG.md with all new features
chore(21/60): add benchmark-full and alerts-check targets to Makefile
refactor(22/60): add --output CLI arg and type annotations to scripts/retrain.py
chore(23/60): bump version to 1.3.0, add SIM/RET/ERA ruff rules, enhance mypy config
feat(24/60): add /api/v1/model/feature-importance endpoint to app/main.py
test(25/60): add TestGetFeatureImportance tests to test_model.py
feat(26/60): add cache_ttl, batch_limit, drift_monitoring settings to config
feat(27/60): add ModelNotReadyError, DriftDetectedError, StationNotFoundError to exceptions
feat(28/60): add get_histogram_samples, get_slowest_requests, clear_metric to telemetry
feat(29/60): add get_logger helper and set_log_level dynamic control to logging_config
refactor(30/60): add precip R2 and sample-count quality gates to retrain_dag
feat(31/60): add RMSE, accuracy, F1 metrics to scripts/evaluate.py
feat(32/60): add argparse CLI args to scripts/load_test.py
feat(33/60): add alembic migration 003 for drift_threshold_log table
test(34/60): update version assertion to 1.3.0 and add TestNewSettings in test_config.py
test(35/60): add TestGetHistogramSamples, TestGetSlowestRequests, TestClearMetric
test(36/60): add tests for ModelNotReadyError, DriftDetectedError, StationNotFoundError
test(37/60): add extreme_weather, cold_weather, reference_values fixtures to conftest.py
feat(38/60): add moving_average_window to app/utils.py
feat(39/60): add validate_coordinate_pair and validate_wind_speed to validators.py
test(40/60): add TestMovingAverageWindow tests to test_utils.py
test(41/60): add TestValidateCoordinatePair and TestValidateWindSpeed to test_validators.py
feat(42/60): add /api/v1/health/detailed and /api/v1/version endpoints to main.py
docs(43/60): update ARCHITECTURE.md with v1.3.0 feature pipeline and new endpoints
docs(44/60): expand DEPLOYMENT.md with new v1.3.0 env vars and health check examples
docs(45/60): add heat index and pressure tendency features + v1.3.0 eval metrics
test(46/60): add TestGetLogger and TestSetLogLevel to test_logging_config.py
test(47/60): add TestHealthDetailed and TestVersionEndpoint to test_api.py
feat(48/60): add get_model_metadata function to app/model.py
feat(49/60): add get_station_prediction_summary to app/database.py
feat(50/60): add get_prediction_stats_by_hour to app/monitoring.py
feat(51/60): add /api/v1/monitoring/stats-by-hour and /api/v1/model/metadata endpoints
test(52/60): add TestStatsByHour and TestModelMetadata tests to test_api.py
```

```
test(53/60): add TestGetModelMetadata tests to test_model.py
test(54/60): add TestGetStationPredictionSummary to test_database.py
test(55/60): add TestGetPredictionStatsByHour to test_monitoring.py
test(56/60): add TestRetrainEndpointExtended for model_version, n_features, r2 checks
docs(57/60): add v1.3.0 new endpoints section to API_REFERENCE.md
feat(58/60): add count_engineered_features helper to app/features.py
test(59/60): add TestCountEngineeredFeatures and TestGetPipelineStageNames to test_features.py
docs(60/60): update README.md with v1.3.0 features, new endpoints, and 8-stage pipeline
```

*Generated by Reflective Lantern | Claude Sonnet 4.6*

*Session: https://claude.ai/code/session_01M5qDQxwsxTqfVwm1Rid6MZ*